

Smooth Control of Remotely Operated Underwater Drones: A Fuzzy-Adaptive Sliding Mode Approach

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Abstract— Controlling underwater robots with nonlinear dynamics is a challenging task. Sliding Mode Control (SMC), a widely used framework for underwater vehicles, suffers from chattering effects that degrade performance. Despite this drawback, SMC remains popular due to its inherent robustness against model uncertainties and external disturbances—an essential requirement in the unpredictable underwater environment. This study proposes a Type-1 fuzzy-adaptive SMC to mitigate this issue, where fuzzy logic dynamically tunes SMC parameters for smoother operation. The method significantly reduces chattering in the baseline SMC model while preserving its robustness. The proposed approach is implemented and validated in the Gazebo simulation platform. This control strategy enhances user experience and vehicle longevity by minimizing high-frequency oscillations, offering practical benefits for remotely operated systems.

Keywords— Fuzzy System, Sliding-Mode Control

I. INTRODUCTION

Underwater drones, also referred to as Remotely Operated Vehicles (ROVs) [1] or tethered unmanned underwater robots, are unoccupied and highly maneuverable systems controlled from a vessel or shore-based station. These systems allow human operators to remain in safe, controlled environments while the drone performs tasks in hazardous underwater conditions. The use of sophisticated remotely controlled systems offers clear advantages for offshore operations, including enhanced safety, reduced costs, and improved data quality [2]. These drones play a crucial role across various sectors by enabling inspection and light intervention missions in environments that are otherwise inaccessible or dangerous, including operations related to offshore infrastructure maintenance, defense, search and rescue, and scientific exploration [3]. Despite their significance, conventional underwater drone control systems demand highly trained field operators, resulting in considerable training and operational costs [4]. Advanced control strategies can address this limitation by enabling more intuitive and efficient vehicle handling.

The control of underwater drones presents a range of complex and persistent challenges. Key factors contributing to this difficulty include the vehicle's highly nonlinear and time-varying dynamics, uncertainties in hydrodynamic coefficients, environmental disturbances such as ocean

currents, and shifting centers of gravity and buoyancy caused by manipulator movements, which further destabilize the system [5]. Traditional controllers like proportional-integral-derivative (PID) are commonly used for their simplicity [6], [7] but lack effectiveness in handling nonlinear dynamics and unmodeled hydrodynamic disturbances [8]. To address these shortcomings, various advanced control techniques have been explored, including model-based linearization methods [9], cascade adaptive control algorithms [10], nonlinear tracking strategies [11], and Sliding Mode Control (SMC) [12]. In more recent developments, intelligent computational techniques such as fuzzy logic and neural networks have gained attention for their potential to enhance robustness [13], [14], [15]. Among these approaches, SMC has emerged as a widely used method due to its inherent robustness and effectiveness in handling both structured and unstructured uncertainties. Its ability to maintain reliable performance in the face of modeling errors and external disturbances makes SMC a compelling choice for underwater drone control [16].

SMC is a nonlinear control strategy that uses a discontinuous control signal to drive system states toward and maintain them on a defined sliding surface. The methodology for applying SMC to underwater vehicles was detailed by Slotine et al. in 1983, where they demonstrated its implementation on a single-input single-output underwater system [17]. Despite its robustness, a major drawback of SMC is the chattering phenomenon—rapid switching in the control input. This behavior can cause practical issues such as overheating in power electronics, accelerated wear in actuators, and the excitation of unmodeled high-frequency dynamics, all of which can impair control performance [18]. To mitigate these effects, a range of chattering reduction strategies have been proposed. These include boundary layer technique [19], disturbance observer-based methods [20], and the use of alternative continuous approximations to the traditional discontinuous sign function [5]. In addition, some studies have incorporated neural networks to suppress chattering while preserving the core robustness of SMC [21].

While various techniques have been proposed to reduce chattering in SMC, there remains a need for more optimized and lightweight solutions. In this study, we present a Fuzzy-based Adaptive Sliding Mode Control (FA-SMC) approach, enhanced using a boundary layer method [19] for chattering reduction. The controller employs a Type-1 fuzzy system to

adaptively tune control parameters, aiming to achieve smoother and more robust performance. The proposed method was implemented and evaluated in the Gazebo simulation environment using a WS Ray, underwater drone developed by the Department of Mechanical Engineering at the University of Moratuwa, Sri Lanka. Section 2 presents the dynamic modeling of the drone. Section 3 discusses the SMC framework and introduces the FA-SMC methodology. Section 4 details the underwater drone model and simulation setup. Section 5 covers simulation results and analysis, followed by conclusions in Section 6.

II. DYNAMIC MODEL OF THE UNDERWATER DRONE

A. Nonlinear Equation of Motion

The motion of an underwater drone is typically described using six degrees of freedom (DOF), represented in two coordinate frames: the body-fixed frame and the earth-fixed frame (Fig. 1). The body-fixed frame is usually defined such that its origin coincides with the vehicle's center of gravity [22].

The vehicle's dynamics can be represented by the following nonlinear equations:

$$\mathbf{M}\dot{\mathbf{v}} + \mathbf{C}(\mathbf{v})\mathbf{v} + \mathbf{D}(\mathbf{v})\mathbf{v} + \mathbf{g}(\boldsymbol{\eta}) + \boldsymbol{\tau}_e = \boldsymbol{\tau} \quad (1)$$

$$\dot{\boldsymbol{\eta}} = \mathbf{J}(\boldsymbol{\eta}_2)\mathbf{v} \quad (2)$$

Equation (1) expresses the nonlinear dynamics of the vehicle, while Equation (2) describes the kinematic relationship between the earth-fixed and body-fixed frames. Both equations are necessary to accurately model and solve the motion of the underwater vehicle.

In these equations, The position and orientation vector in the earth fixed frame is denoted as $\boldsymbol{\eta} = [x, y, z, \phi, \theta, \psi]^T$, which can be partitioned into $\boldsymbol{\eta}_1$ and $\boldsymbol{\eta}_2$. The body fixed velocity vector $\mathbf{v} = [u, v, w, p, q, r]^T$ includes linear and angular components.

The system dynamics involve the symmetric positive-definite inertia matrix $\mathbf{M}$, combining rigid-body and added mass effects. The Coriolis and centripetal matrix $\mathbf{C}$ —dependent on the body-fixed velocity vector $\mathbf{v}$ —models the nonlinear velocity-dependent forces. Hydrodynamic drag and lift forces are modeled by damping matrix $\mathbf{D}$, simplified to diagonal linear and quadratic terms, assuming uncoupled motion and neglecting higher-order effects. Restoring forces from gravity and buoyancy are represented by $\mathbf{g}$, while the transformation matrix $\mathbf{J}$ maps body-fixed velocities to earth-fixed derivatives. The dynamic model is completed by control input $\boldsymbol{\tau}$ and external disturbances $\boldsymbol{\tau}_e$ which account for environmental interactions such as current forces and tether dynamics.

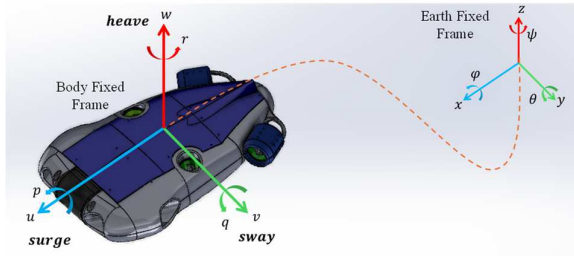


Figure 1. Earth-fixed and body fixed frames, and relating transformation for the WS Ray drone configuration.

For modeling purposes, (1) can also be expressed in earth-fixed coordinates, as shown below to support the control design process [5].

$$\mathbf{M}_{\boldsymbol{\eta}}(\boldsymbol{\eta})\ddot{\boldsymbol{\eta}} + \mathbf{C}_{\boldsymbol{\eta}}(\boldsymbol{\eta}, \dot{\boldsymbol{\eta}})\dot{\boldsymbol{\eta}} + \mathbf{D}_{\boldsymbol{\eta}}(\boldsymbol{\eta}, \dot{\boldsymbol{\eta}})\dot{\boldsymbol{\eta}} + \mathbf{g}_{\boldsymbol{\eta}}(\boldsymbol{\eta}) + \boldsymbol{\tau}_{e_{\boldsymbol{\eta}}} = \boldsymbol{\tau}_{\boldsymbol{\eta}} \quad (3)$$

where,

$$\begin{aligned} \mathbf{M}_{\boldsymbol{\eta}}(\boldsymbol{\eta}) &= \mathbf{J}^{-T}(\boldsymbol{\eta})\mathbf{M}\mathbf{J}^{-1}(\boldsymbol{\eta}), \\ \mathbf{C}_{\boldsymbol{\eta}}(\boldsymbol{\eta}, \dot{\boldsymbol{\eta}}) &= \mathbf{J}^{-T}(\boldsymbol{\eta})[\mathbf{C}(\mathbf{v}) - \mathbf{M}\mathbf{J}^{-1}(\boldsymbol{\eta})\dot{\mathbf{J}}(\boldsymbol{\eta})]\mathbf{J}^{-1}(\boldsymbol{\eta}), \\ \mathbf{D}_{\boldsymbol{\eta}}(\boldsymbol{\eta}, \dot{\boldsymbol{\eta}}) &= \mathbf{J}^{-T}(\boldsymbol{\eta})\mathbf{D}(\mathbf{v})\mathbf{J}^{-1}(\boldsymbol{\eta}), \\ \mathbf{g}_{\boldsymbol{\eta}}(\boldsymbol{\eta}) &= \mathbf{J}^{-T}(\boldsymbol{\eta})\mathbf{g}(\boldsymbol{\eta}), \text{ and } \boldsymbol{\tau}_{e_{\boldsymbol{\eta}}}(\boldsymbol{\eta}) = \mathbf{J}^{-T}(\boldsymbol{\eta})\boldsymbol{\tau}_e \end{aligned} \quad (4)$$

B. Thruster Model

The thrust forces $\mathbf{u}$ represent the output vector of the thruster system, whose dynamics are inherently nonlinear and complex. To simplify the control and modeling process, an affine thrust model is typically used, in which the system is considered linear with respect to its inputs [22].

$$\boldsymbol{\tau} = \mathbf{B}\mathbf{u} \quad (5)$$

Here, $\boldsymbol{\tau}$ denotes the control input vector applied to the vehicle, and $\mathbf{u}$ is the vector of individual thruster forces. The matrix $\mathbf{B} \in \mathbb{R}^{6 \times p}$, commonly referred to as the thrust configuration matrix, defines the mapping from the thruster forces to the generalized forces and moments acting on the vehicle. This matrix is constant and depends on the geometric configuration of the thrusters. Equation (5) is often used to compute the required thrust values for each individual propeller, given a desired control input $\boldsymbol{\tau}$.

III. SLIDING MODE CONTROL AND PROPOSED FUZZY-ADAPTIVE APPROACH

A. Sliding Mode Control: Principles and Formulation

SMC is a model-based control strategy that leverages the system's dynamic equations to both predict its behavior and correct deviations from the desired trajectory. Its formulation is particularly well-suited for handling nonlinearities, velocity-dependent forces, and the coupled dynamics commonly found in underwater drones [5].

For multivariable systems such as underwater drones, the sliding surface must be appropriately defined to drive the tracking error to zero. The general sliding surface can be expressed as:

$$\mathbf{s} = \dot{\mathbf{e}} + \lambda\mathbf{e} \quad (6)$$

Where $\mathbf{s} \in \mathbb{R}^6$ is the sliding variable, $\mathbf{e}$ is the tracking error, and $\lambda > 0$ is a design parameter that determines the control bandwidth.

For controller synthesis, it is convenient to introduce a virtual reference $\boldsymbol{\eta}_r$ such that:

$$\dot{\boldsymbol{\eta}}_r = \dot{\boldsymbol{\eta}}_d + \lambda\mathbf{e} \rightarrow \mathbf{s} = \dot{\boldsymbol{\eta}} - \dot{\boldsymbol{\eta}}_r \quad (7)$$

Maintaining the system trajectory on the sliding surface ensures convergence of the actual state $\boldsymbol{\eta}(t)$ to the desired trajectory $\boldsymbol{\eta}_d(t)$ over time [22].

To derive the control law, a Lyapunov-like candidate function is considered:

$$V(\mathbf{s}, t) = \frac{1}{2}\mathbf{s}^T\mathbf{M}_0(\boldsymbol{\eta})\mathbf{s} \quad (8)$$

Differentiating $V(s, t)$ with respect to time and using the sliding surface definition (7) yields:

$$\begin{aligned}\dot{V}(s, t) &= \mathbf{s}^T \mathbf{M}_0(\boldsymbol{\eta}) \dot{\mathbf{s}} + \frac{1}{2} \mathbf{s}^T \dot{\mathbf{M}}_0(\boldsymbol{\eta}) \mathbf{s} \\ &= \mathbf{s}^T \mathbf{M}_0(\boldsymbol{\eta}) (\ddot{\boldsymbol{\eta}} - \ddot{\boldsymbol{\eta}}_r) + \frac{1}{2} \mathbf{s}^T \dot{\mathbf{M}}_0(\boldsymbol{\eta}) \mathbf{s}\end{aligned}\quad (9)$$

Substituting $\mathbf{M}_0(\boldsymbol{\eta}) \ddot{\boldsymbol{\eta}}$ from (3) into (9) and using the skew-symmetric property of $\left[\frac{1}{2} \mathbf{M}_0(\boldsymbol{\eta}) - \mathbf{C}_0(\mathbf{v}, \boldsymbol{\eta})\right]$, we obtain:

$$\dot{V} = \mathbf{s}^T [\boldsymbol{\tau} - \mathbf{C}_0(\mathbf{v}, \boldsymbol{\eta}) \dot{\boldsymbol{\eta}}_r - \mathbf{D}_0(\mathbf{v}, \boldsymbol{\eta}) \dot{\boldsymbol{\eta}} - \mathbf{g}_0(\boldsymbol{\eta}) - \boldsymbol{\delta} - \mathbf{M}_0(\boldsymbol{\eta}) \ddot{\boldsymbol{\eta}}_r]\quad (10)$$

Here $\boldsymbol{\delta}$ represents all the uncertain terms and environmental disturbances. To ensure the $\dot{V} < -\epsilon \|\mathbf{s}\|$, where $\epsilon > 0$, the control input $\boldsymbol{\tau}$ can be designed as:

$$\boldsymbol{\tau} = \boldsymbol{\tau}_{eq} + \boldsymbol{\tau}_{dis}\quad (11)$$

with

$$\boldsymbol{\tau}_{eq} = \mathbf{M}_0(\boldsymbol{\eta}) \ddot{\boldsymbol{\eta}}_r + \mathbf{C}_0(\mathbf{v}, \boldsymbol{\eta}) \dot{\boldsymbol{\eta}}_r + \mathbf{D}_0(\mathbf{v}, \boldsymbol{\eta}) \dot{\boldsymbol{\eta}} - \mathbf{g}_0(\boldsymbol{\eta})\quad (12)$$

$$\boldsymbol{\tau}_{dis} = -K \text{sgn}(\mathbf{s})\quad (13)$$

Here $\boldsymbol{\tau}_{eq}$ is the equivalent control and $\boldsymbol{\tau}_{dis}$ is a discontinuous term used to reject uncertainties and disturbances.

B. Fuzzy-Adaptive Sliding Mode Control (FA-SMC)

To mitigate chattering—one of the primary drawbacks in SMC—Yoeager et al. introduced a boundary layer approach by replacing the discontinuous switching function $\text{sgn}(\mathbf{s})$ in the discontinuous term $\boldsymbol{\tau}_{dis}$ in (13) with a continuous switching function, $\text{sat}(\mathbf{s}/\alpha)$, where α denotes the boundary layer thickness [19]. However, the convergence behavior of the system is not solely governed by the boundary layer. Theoretically, the gain parameter K , especially within the equivalent control term, significantly influences the rate at which the system converges to the sliding surface. While a higher K value improves convergence speed, it can also intensify chattering by inducing sharp control actions. Similarly, the parameter λ in the sliding surface formulation modulates the relative influence of the position error. A larger λ increases the contribution of the position error in the sliding surface, leading to more aggressive corrective action and faster convergence, but with a higher risk of overshoot or instability.

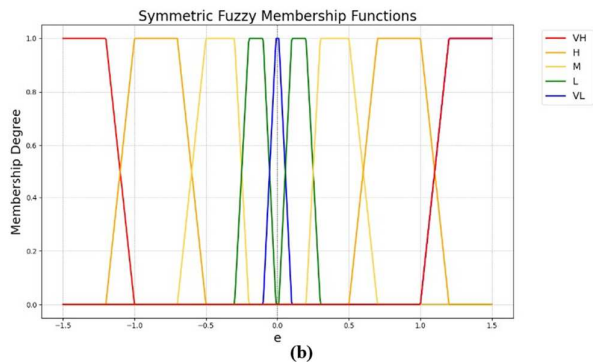
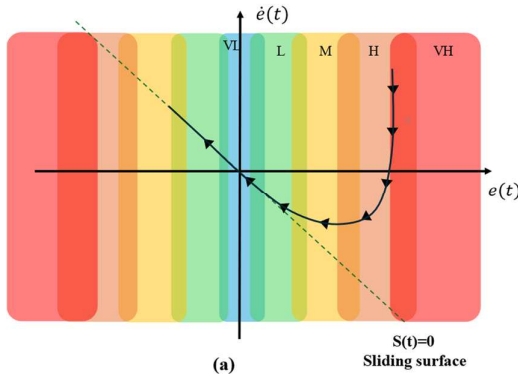


Figure 2. (a) Graphical representation of the sliding mode control (SMC) approach, illustrating the sliding surface and fuzzy linguistic regions to facilitate smooth control. (b) Trapezoidal membership functions corresponding to the fuzzy linguistic variables, used to adapt control gains based on the system's error state. here (VL-Very Low, L-Low, M-Medium, H-High, and VH-Very High).

Under sudden disturbances, achieving the sliding surface rapidly requires larger values of the disturbance rejection gain K and the surface slope λ , which boost responsiveness and convergence speed. However, these higher gains also sharpen the discontinuities in the control input as the system approaches the surface, leading to chattering. To balance speed and smoothness, an adaptive mechanism is needed that applies greater control effort when the error is large and reduces it near the surface. In this work, we employ a Type-1 fuzzy system to adjust both K and λ based on the magnitude of the error term e , assigning higher values for large deviations and lower values as e approaches zero, thereby ensuring fast convergence while significantly reducing chattering and enhancing overall stability.

The fuzzy system employs trapezoidal membership functions to convert the error term e into linguistic variables (Fig. 2), which are then processed using a weighted average defuzzification strategy. The resulting crisp output is scaled by predefined maximum values for both the K and a λ that bounds the control effort. Each of the six DOFs is assigned a unique gain to account for their distinct dynamic characteristics, particularly the differences between translational and rotational motions. These scaled parameters are then used within the control law to compute the required force vector $\boldsymbol{\tau}$, as given in (11). This approach allows the controller to adapt its response characteristics in real time, improving convergence while maintaining robustness under varying environmental and dynamic conditions.

C. Integration of FA-SMC in Underwater Drone Control

In the context of a remotely operated underwater drone, the control objective is typically defined in terms of velocity commands issued by the user. During real-time operation, the control loop executes periodically, and at each iteration, the drone's current position is treated as the origin of a locally earth-fixed reference frame. This effectively redefines the world frame at each timestep, allowing the user-defined velocity inputs to be interpreted directly as $\dot{\boldsymbol{\eta}}_d$, which are used to compute the sliding variable $\mathbf{s}$ using (7). Based on the magnitude of the tracking error e , the fuzzy system then produces outputs that adapt both K and λ within the control law. These modified parameters are substituted into equivalent control and disturbance rejection terms, as defined in equations (12) and (13). The final control vector $\boldsymbol{\tau}$ is computed accordingly.

The control vector τ is transformed into individual thruster commands using the thrust configuration matrix B via the relationship (5), where u represents the vector of thrust forces. These calculated thrust values are then allocated to the corresponding onboard thrusters. This architecture (Fig. 3) is implemented and tested within a Gazebo-based simulation environment, replicating realistic underwater dynamics for the custom-built underwater drone, WS Ray.

IV. SYSTEM MODELING AND GAZEBO SIMULATION SETUP

The simulation and control framework for the underwater drone was developed using ROS 2 Jazzy and the Gazebo Harmonic simulator, which offers a realistic physics environment for underwater robotics. The dynamic model of the custom-built vehicle, WS Ray, was integrated using a detailed SDF description. Controller feedback was obtained via the Gazebo Odometry plugin.

WS Ray features six thrusters in a conventional layout enabling 5-DOF actuation, with the pitch axis left uncontrollable due to thruster configuration. To reflect real-world limits, control inputs were capped at 20 N per propeller based on actual thruster specs. A fluid density of 1000 kg/m³ was used, representing freshwater or shallow sea conditions.

Key hydrodynamic parameters were derived via ANSYS Fluent and applied in both the controller and simulation setup. These parameters are summarized in Table I.

TABLE I. WS RAY DIMENSIONS AND HYDRODYNAMIC PARAMETERS

Parameters	Nomenclature	Value
Weight	W	137.2 N
Buoyancy	B	144.6 N
Rotational Inertia	I_x, I_y, I_z	0.37m, 0.52, 0.18 kgm ²
Added mass Coeff.	X_u, Y_v, Z_w	4.79, 12.20, 61.56 kg
	K_p, M_q, N_r	6.18, 5.14, 1.44 kgm ²
Linear Drag Coeff.	X_u, Y_v, Z_w	0.161, 0.515, 0.003 kg/s
	K_p, M_q, N_r	0.018, 0.072, 0.032 kgm ² /s
Quadratic Drag Coeff.	$X_u u , Y_v v , Z_w w $	9.58, 12.27, 108.50 Ns ² /m ²
	$K_p p , M_q q , N_r r $	0.003, 0.215, 0.145 Nms ²
Centre of Buoy. (CoB)	r_B	[0, 0, 0.016] m
Maximum Thrust	T_{max}	20 N

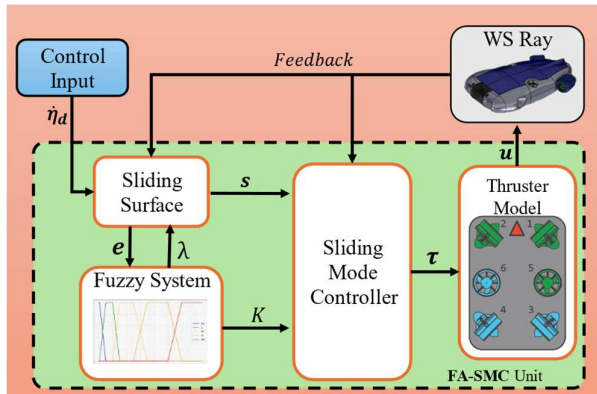


Figure 3. Overview of the FA-SMC control architecture.

V. SIMULATION RESULTS

To evaluate the effectiveness of the proposed FA-SMC, a series of simulation tests were conducted in comparison with conventional SMC and the Bounded SMC technique (B-SMC). All tests were carried out using the WS Ray underwater drone model. For SMC, the gain and slope values were set as $K = [10, 5, 5, 0.5, 1, 2]$ and $\lambda = [5, 5, 5, 1, 1, 1]$. Further increasing either K or λ beyond these values led to unstable behavior, even under still water conditions. In B-SMC, $K = [15, 15, 5, 0.5, 1, 2]$ with the same λ , and a saturation function with $\alpha = 0.07$ was used to mitigate chattering. For FA-SMC, adaptive bounds were set as $\lambda_{max} = [10, 10, 5, 1, 1, 1]$, $K_{max} = [15, 15, 5, 0.5, 1, 2]$, enabling aggressive control under large errors while reducing gains near the sliding surface to smooth the response. These values are also based on two key considerations: (1) underwater vehicles generally demand stronger control in translational degrees of freedom due to their greater inertia and exposure to environmental disturbances, and (2) with the drone's center of buoyancy located above its center of gravity, sudden heave accelerations can induce significant roll and pitch responses. Consequently, using lower gains for the heave axis and its coupled roll-pitch axes helps preserve stability and prevents destabilizing oscillations.

A. Speed Tracking Performance in Still Water

In the first experiment, the underwater drone was commanded to reach a forward velocity of 0.5 m/s, a value representative of typical operational speeds. Both SMC and B-SMC successfully drove the vehicle to the target speed within approximately 0.2 seconds, while FA-SMC achieved the target within 0.25 seconds (Fig. 4).

After reaching the desired velocity, small fluctuations were observed in the velocity profile: around ± 0.05 m/s for SMC and B-SMC, and a significantly smaller fluctuation of ± 0.02 m/s for FA-SMC. Notably, the SMC-controlled drone exhibited mild oscillatory behavior in non-commanded directions (e.g., lateral or rotational drift), indicating cross-coupling effects and lack of smooth convergence. In contrast, B-SMC and FA-SMC achieved the commanded velocity with minimal chattering and no significant oscillations in other degrees of freedom.

These results demonstrate that both B-SMC and FA-SMC provide improved velocity tracking performance and smoother stabilization.

B. Response to Sudden Displacement

To evaluate controller robustness against abrupt external disturbances, a sudden displacement of $[0, 0.5, -0.5, 0, 0, 0.17]$ was introduced while the vehicle was cruising at 0.5 m/s.

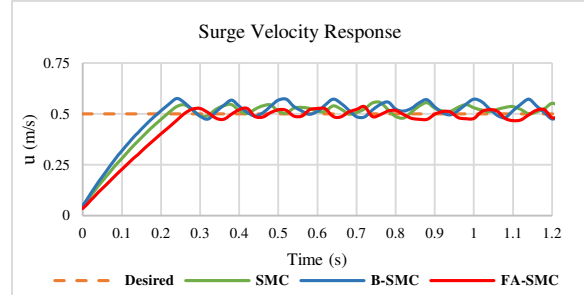


Figure 4. Forward velocity response under SMC, B-SMC, and FA-SMC during target speed tracking.

Under both SMC and B-SMC, the system exhibited significant deviations and failed to fully reconverge to the desired sliding surface within the 10-second window. In contrast, the FA-SMC controller successfully rejected the disturbance and achieved full re-stabilization within this period, demonstrating superior convergence characteristics with minimal corrective transients.

Under FA-SMC, the system maintained or rapidly restored stable behavior across key dynamic axes—particularly surge, pitch, roll, and yaw. While minor transients appeared immediately after the disturbance, full recovery was achieved within 5 seconds, highlighting the controller's adaptability. In comparison, B-SMC occasionally reached the sliding surface faster but failed to maintain consistent convergence, exhibiting lingering oscillations. This was primarily due to the high λ values used in the tracking error term, which, despite the presence of a boundary

layer, introduced residual chattering. These results underscore the advantage of incorporating fuzzy adaptation, allowing dynamic gain adjustment that enhances robustness and smooth stabilization under sudden perturbations.

C. Spiral Trajectory Tracking Performance

To further assess the controller's performance in complex maneuvers, the underwater drone was commanded to follow an upward spiral trajectory (Fig. 6). This path required simultaneous translational and rotational coordination in multiple degrees of freedom. The FA-SMC method demonstrated effective trajectory tracking, maintaining the desired helical path with high stability. Pitch and roll oscillations remained within ± 0.05 rad, indicating strong orientation control even under dynamic multi-axis motion. These results validate the controller's ability to handle more advanced motion profiles beyond linear navigation.

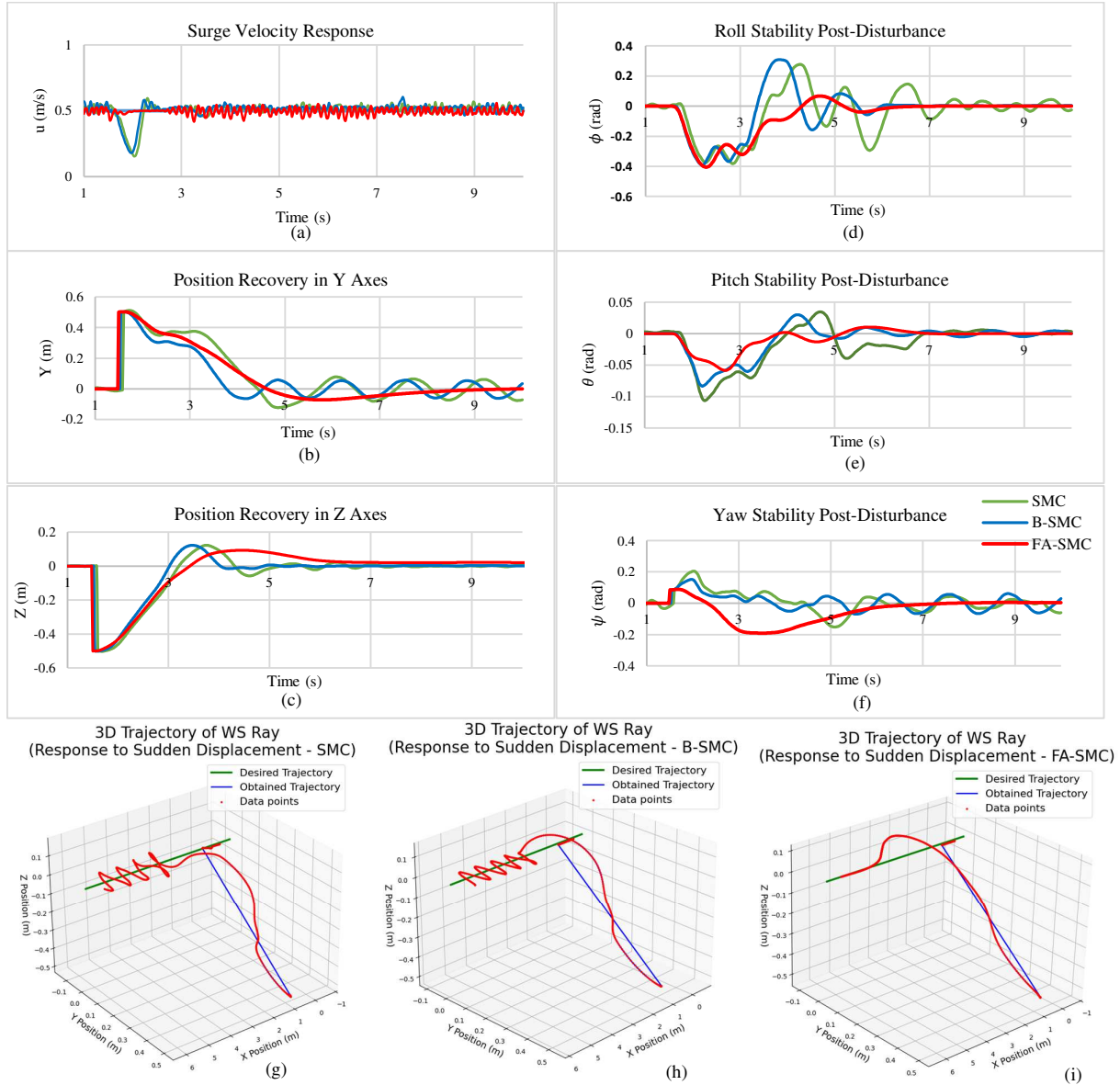


Figure 5. Time series plots of surge velocity, Y–Z position, and orientation recovery (a–f). Green lines represent SMC, blue for B-SMC, and red for FA-SMC. Trajectory responses for SMC (g), B-SMC (h), and FA-SMC (i) during sudden disturbance illustrate the smoother recovery path achieved by FA-SMC.

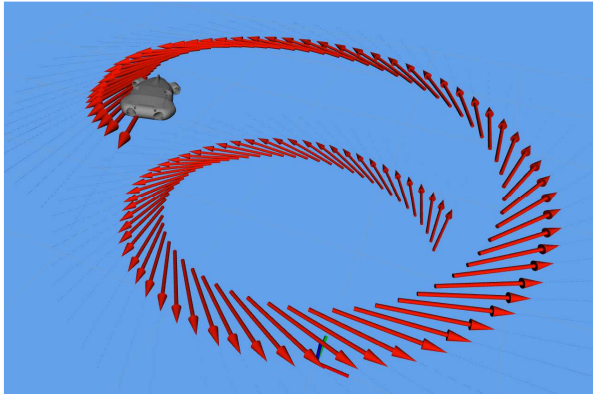


Figure 6. RViz-based visualization during upward spiral trajectory tracking.

VI. CONCLUSION

This paper presented an FA-SMC strategy for enhancing the smoothness and robustness of remotely operated underwater drones. By integrating a Type-1 fuzzy logic system with traditional SMC, the proposed method adaptively modulates both the disturbance rejection gain (K) and surface slope (λ) based on real-time error trends. This significantly reduces chattering while maintaining strong convergence properties.

The control approach was implemented on the WS Ray underwater drone within a ROS 2 Jazzy and Gazebo Harmonic simulation environment. While traditional SMC methods exhibited residual chattering and limited adaptability, the B-SMC method also struggled to handle abrupt disturbances effectively and required careful tuning of parameters for optimal performance. In contrast, the FA-SMC method demonstrated robust behavior with minimal parameter adjustment, maintaining stability even under sudden displacements. This adaptive mechanism not only enhances resilience but also enables smoother control, aligning with the objective of achieving refined motion in underwater drone operations.

As a future direction, this controller will be validated on a physical prototype to evaluate sim-to-real transfer performance. Additional testing—such as station-keeping under wave disturbances and operation in ocean current conditions—will be essential to assess the system's viability for real-world applications, particularly in scenarios like coral reef observation where precise and stable positioning is critical.

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